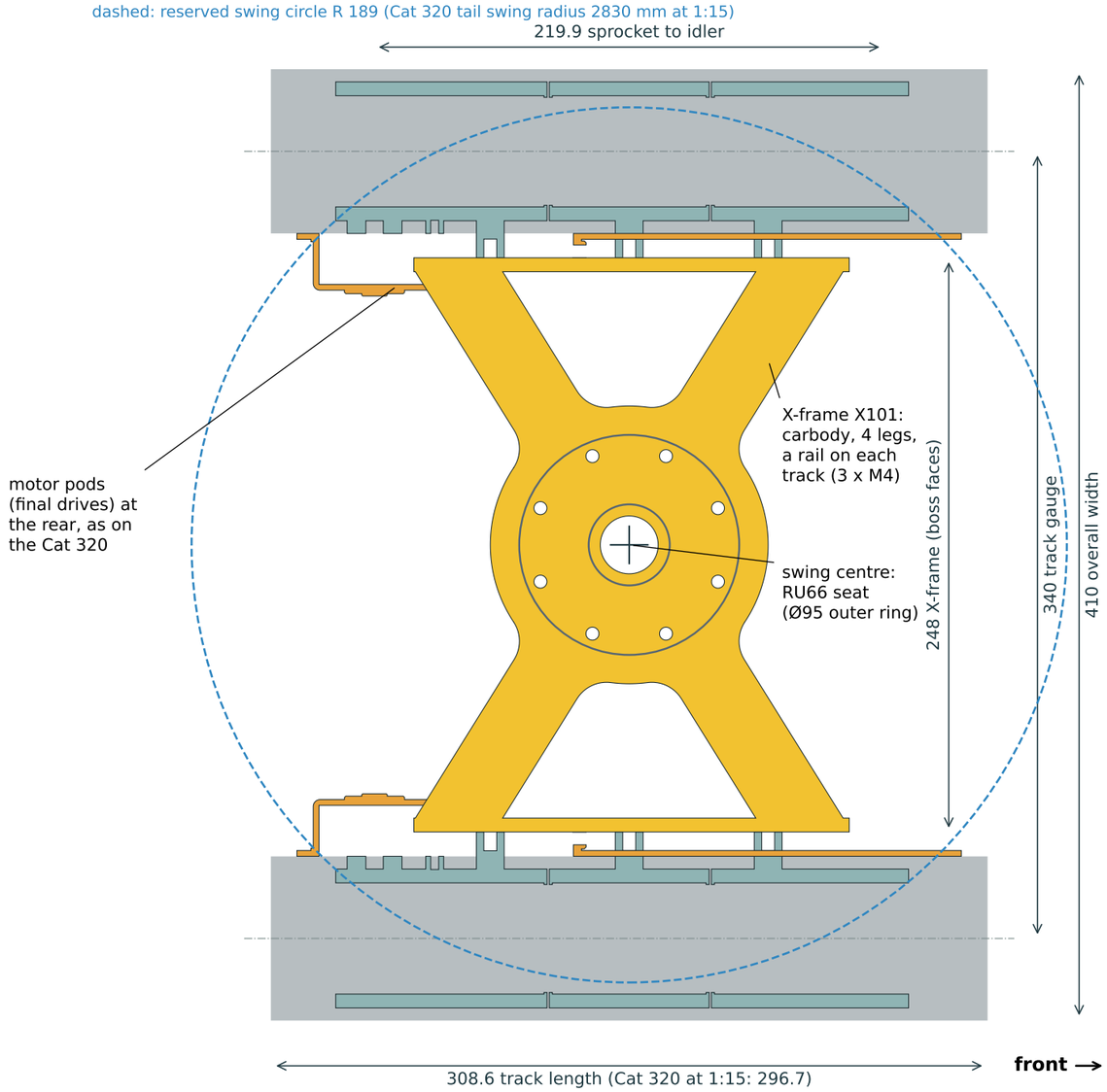
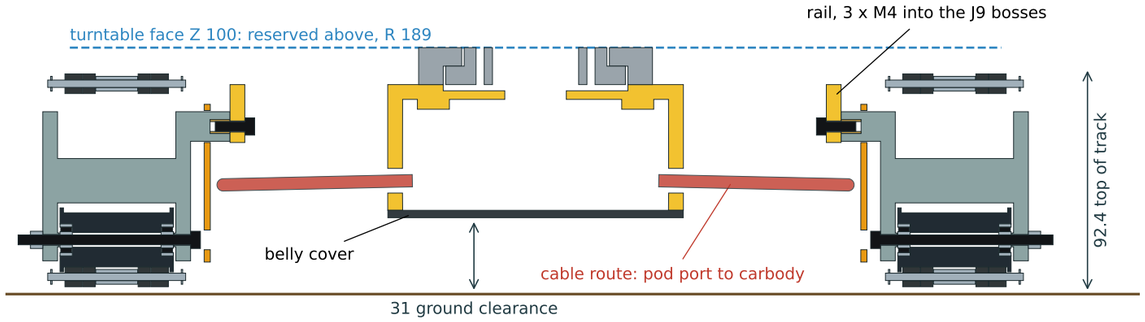
X1 crawler chassis — two J9 tracks on an X-frame (built from the CAD data)

Cat 320 lower-frame layout: round carbody under the slewing ring, four legs in an X, final drives at the rear. The turntable is not built; its seat, cable passage and swing envelope are reserved.

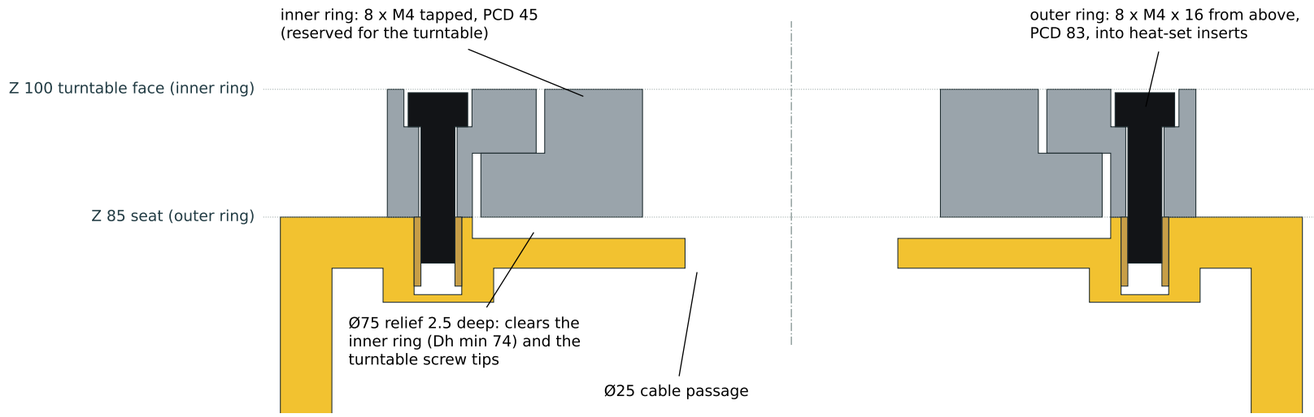
Plan view (front to the right). Sections: X-frame at Z 80, J9 frames at Z 68, motor pods at the axle height



Front section through the swing centre (X = 0), seen from the front



Turntable seat: section through a seat screw (THK RU66, catalogue A18-21)



Cat 320 reference (07H spec sheet p. 4) and X1 (mm)

Item	Cat 320	1:15	X1
Track length	4,450	296.7	308.6
Length to centre of rollers	3,650	243.3	219.9
Track gauge	2,380	158.7	340.0
Undercarriage width	3,170	211.3	410.0
Shoe width	790	52.7	70.0
Ground clearance	470	31.3	31.0
Counterweight clearance / turntable face	1,050	70.0	100.0
Tail swing radius / reserve	2,830	188.7	189.0
Track / carrier rollers per side	8 / 2		5 / 2

CAD checks (validation.json): PASS

static: 0 overlaps in 126 exact pairs (both tracks, X-frame, RU66, screws, cable routes)

turntable reserve (R 189 above Z 100): no intrusion; tracks 7.6 below it, X-frame 15

track slide-on (both sides), RU66 lowering, belly cover: no overlaps; 46 tool paths clear

support-roller bolts: unbolt the track from the X-frame before withdrawing them

Dimensions in mm. Vehicle frame: X forward, Y left, Z up, origin on the ground under the swing centre.
The right track is the mirror image of J9 (mirrored J901 and J909 prints).